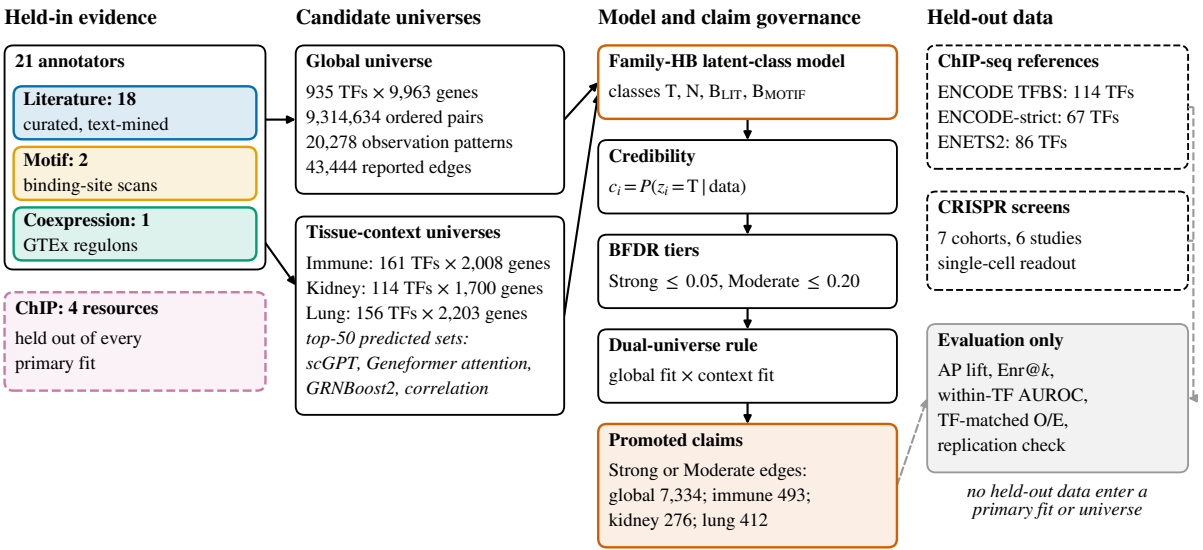
Held-in evidence
21 annotators
Literature: 18
curated, text-mined
Motif: 2
binding-site scans
Coexpression: 1
GTEx regulons
ChIP: 4 resources
held out of every
primary fit
Candidate universes
Global universe
935 TFs × 9,963 genes
9,314,634 ordered pairs
20,278 observation patterns
43,444 reported edges
Tissue-context universes
Immune: 161 TFs × 2,008 genes
Kidney: 114 TFs × 1,700 genes
Lung: 156 TFs × 2,203 genes
top-50 predicted sets:
scGPT, Geneformer attention,
GRNBoost2, correlation
Model and claim governance
Family-HB latent-class model
classes T, N, B_LIT, B_MOTIF
Credibility
c_i = P(z_i = T | data)
BFDR tiers
Strong ≤ 0.05, Moderate ≤ 0.20
Dual-universe rule
global fit × context fit
Promoted claims
Strong or Moderate edges:
global 7,334; immune 493;
kidney 276; lung 412
Held-out data
ChIP-seq references
ENCODE TFBS: 114 TFs
ENCODE-strict: 67 TFs
ENETS2: 86 TFs
CRISPR screens
7 cohorts, 6 studies
single-cell readout
Evaluation only
AP lift, Enr@k,
within-TF AUROC,
TF-matched O/E,
replication check
no held-out data enter a
primary fit or universe